When Standard Retrieval Optimizations Fail: A Cost-Aware Ablation Study of RAG Pipelines for Financial Document Question Answering

Anonymous Author(s)[1]

[1]Anonymous Institution(s)

Submitted for double-blind review
DRAFT — June 2026 | *Target: WSDM 2027*

Abstract

We present a controlled ablation study testing five standard RAG retrieval improvements — semantic chunking, recursive chunking, cross-encoder reranking, metadata enrichment, and HyDE query expansion — on the FinanceBench benchmark of financial document question answering, evaluated across 133 questions from 10-K, 10-Q, and earnings transcript filings. The central finding is negative and informative: none of the five techniques improve meaningfully over a fixed 512-token chunking baseline on ROUGE-L F1. We show this is not a failure of implementation but a consequence of structural chunking failure — fixed-token splitting destroys table structure in SEC filings, and no retrieval technique can recover passages that were never coherently chunked. A secondary finding corrects an evaluation methodology problem: ROUGE-L F1 understates model performance on numeric questions, where Exact Match reveals 27.7–31.9% correct numerical answers under the best conditions despite near-zero ROUGE-L scores — a 10–13x discrepancy. We report the negative result directly, enumerate limitations explicitly, and derive from the results a practical recommendation: invest in table-aware document parsing before any retrieval optimization. Full code and results will be made available at a public repository upon acceptance (link omitted for anonymous review).

Keywords: retrieval-augmented generation, financial document QA, chunking, reranking, FinanceBench, SEC filings, negative results, cost-accuracy tradeoff

1 Introduction

Enterprise question answering over SEC filings is a commercially important problem that standard RAG pipelines solve poorly. The failure rate is documented and large: Islam et al. [1] show that GPT-4-Turbo with a standard retrieval system incorrectly answers or refuses to answer 81% of questions on FinanceBench even when the answer is present in the document. Our own C0 baseline — GPT-4o with fixed 512-token chunking and cosine similarity retrieval — achieves mean ROUGE-L F1 = 0.177 across 133 questions, and 0.028 on metrics-generated numeric questions specifically, confirming near-total failure on the hardest question type.

The obvious response to such failure rates is to deploy standard RAG improvements: better chunking, reranking, metadata filtering, query expansion. This paper tests whether that response works on financial documents. It does not — and understanding why is the contribution.

This paper makes the following contributions, scoped to what is implemented and measured:

- A controlled ablation across six RAG pipeline configurations on FinanceBench (C0–C5), with each technique varied independently to isolate its effect. C1 (semantic chunking) and C6 (hybrid) are pending a software dependency resolution.
- A negative result with explanatory power: no tested technique improves ROUGE-L F1 over the C0 baseline. The failure is structural — table contents are destroyed during text splitting — and no retrieval technique recovers passages that were never coherently chunked.
- A correction to evaluation methodology: ROUGE-L F1 understates model performance on numeric questions by 10–13x. Exact Match reveals 27.7–31.9% correct numerical answers under the best conditions despite near-zero ROUGE-L scores.
- A cost-efficiency finding: since all techniques achieve approximately equal ROUGE-L F1, the cheapest configuration is also the most efficient. Reranking adds 2.9x latency at no accuracy gain.

We report the negative result directly. A paper that applied techniques known to work on general-domain benchmarks to financial documents without reporting failure would misrepresent the problem.

1.1 What Is and Is Not in This Paper

What is implemented and measured: a six-condition ablation on 133 FinanceBench questions using GPT-4o (temperature=0) and all-MiniLM-L6-v2 embeddings, single-run estimates.

What is not in this paper: three-seed replication (single-run only; the immediate next step), cost-per-query in USD (OpenAI usage logs available but not yet aggregated), C1 and C6 results (LangChain 1.3.11 import dependency conflict), table-aware chunking as a tested condition (identified as future work by the results).

2 Related Work

2.1 Financial Document QA

FinanceBench [1] provides the benchmark: 10,231 questions over SEC filings, with GPT-4-Turbo failing 81% under standard retrieval. FinQA [4] established the difficulty of multi-step numerical reasoning over financial tables. FinAgentBench [5] treats each table as a single chunk unit — a design directly motivated by the table-structure problem we identify. Greenback Bears [10] shows 19% accuracy with ada-002 embeddings alone on FinanceBench.

2.2 RAG Optimizations on Financial Documents

Vidra et al. [2] test chunking, query expansion, metadata, and reranking on FinanceBench — the direct predecessor. Our work extends theirs with a fully controlled ablation (each technique varied independently), a negative result not present in their paper, and cost and latency measurement. Patel et al. [3] find that BM25

Table 1: Ablation conditions. C0 is the baseline; each of C1–C5 varies exactly one component.

Condition	Chunking	Rerank.	Metadata	Q. Exp.
C0 — Baseline	Fixed 512 tok., 50-tok. overlap	—	—	—
C1 — Semantic (pending)	Semantic boundary	—	—	—
C2 — Recursive	Recursive, 2,000-tok. max	—	—	—
C3 — + Reranking	Fixed 512 tokens	BGE-v2-m3	—	—
C4 — + Metadata	Fixed 512 tokens	—	Co./Yr./Type prefix	—
C5 — + Query Exp.	Fixed 512 tokens	—	—	HyDE
C6 — Hybrid (pending)	Recursive	BGE-v2-m3	Prefix	HyDE

outperforms dense retrieval and that query expansion provides limited benefit for precise numeric queries — consistent with our C5 finding. Banerjee et al. [7] focus on metadata-driven RAG (relevant to C4). Kim et al. [8] demonstrate multi-reranker ensembling (relevant to C3).

2.3 The Metric Problem

ROUGE-L F1 is the standard FinanceBench metric. Section 5.2 shows it understates model capability on numeric questions by 10–13x. This is not noted in prior work. We recommend reporting Exact Match alongside ROUGE-L for financial QA benchmarks.

3 Methodology

3.1 Dataset

We evaluate on the publicly available FinanceBench split [1] — 150 questions spanning metrics-generated (47 questions; multi-step numeric calculations), novel-generated (40 questions; extractive and interpretive), and domain-relevant (46 questions; qualitative analysis). We use 133 of 150 questions after excluding 17 whose source PDFs could not be downloaded: Adobe 10-Ks (2015, 2016, 2017, 2022), Johnson & Johnson (2022 10-K, Q4 2022 and Q2 2023 earnings, 2023 8-K), and MGM Resorts (Q4 2022 earnings). This exclusion is not random; company pages that block automated downloads may differ systematically in document structure.

3.2 Ablation Design

Each of C1–C5 varies exactly one component from C0. C6 combines all techniques and is pending.

Table 2: C0 Baseline by question type. Note the 10x ROUGE-L gap between metrics-generated and qualitative types, and the Exact Match / ROUGE-L discrepancy on metrics-generated (§5.2).

Question Type	N	ROUGE-L	EM	Note
metrics-generated	47	0.028	0.277	13/47 exact; near-0 ROUGE-L
novel-generated	40	0.261	0.050	Prose retrieval works
domain-relevant	46	0.257	0.022	Qualitative retrieved well
All	133	0.177	0.128	10x F1 gap across types

3.3 Components

C0 — Fixed Chunking (Baseline). TokenTextSplitter, chunk_size=512, chunk_overlap=50. The most widely used default RAG chunking strategy. All other conditions are measured relative to this.

C2 — Recursive Chunking. RecursiveCharacterTextSplitter with separator hierarchy (section headers, paragraphs, lines, sentences, words, characters) and chunk_size=2,000. Intended to preserve table rows by using larger context windows.

C3 — Reranking. Base retriever fetches k=20 chunks via cosine similarity. BAAI/bge-reranker-v2-m3 re-scores each as a cross-encoder and retains top-5 for generation. Adds 23 seconds mean latency per query on the hardware used.

C4 — Metadata Enrichment. Each chunk annotated with company ticker, fiscal year, and filing type, prepended as a structured prefix. Retrieval query similarly enriched with the same fields parsed from the doc_name column.

C5 — Query Expansion (HyDE). GPT-4o generates a hypothetical analyst answer passage per question. This passage and the original query are each embedded; results are merged (deduplicated on first-100-character prefix) and top-k unique chunks passed to generation. Adds one extra GPT-4o call per query.

3.4 Metrics

ROUGE-L F1 (standard FinanceBench metric; primary accuracy signal), Exact Match (gold answer appears as substring of prediction; added because Section 5.2 shows ROUGE-L understates numeric performance), and mean wall-clock latency in seconds. Cost-per-query will be added in the next revision.

3.5 Implementation

GPT-4o (temperature=0) for generation; all-MiniLM-L6-v2 for embedding; FAISS for vector search; k=10 across all conditions; Python 3.11, LangChain 1.3.11. Single-run results. All code and results are in the repository.

4 Results

4.1 C0 Baseline

Mean ROUGE-L F1 = 0.177, Exact Match = 0.128, mean latency = 12.3s across 133 questions.

4.2 Full Ablation Results

We report the headline result plainly: no measured condition improves ROUGE-L F1 over C0.

Table 3: Full ablation results. All measured conditions perform at or below C0 on ROUGE-L F1. Negative deltas indicate degradation. C1 and C6 pending.

Condition	ROUGE-L	EM	Lat. (s)	Δ vs. C0
C0 — Baseline	0.177	0.128	12.3	—
C1 — Semantic (pending)	—	—	—	—
C2 — Recursive	0.177	0.120	11.2	0.000
C3 — + Reranking	0.175	0.128	35.3	−0.002
C4 — + Metadata	0.170	0.105	10.8	−0.007
C5 — + Query Exp.	0.167	0.098	13.2	−0.010
C6 — Hybrid (pending)	—	—	—	—

Table 4: ROUGE-L F1 by question type, all measured conditions. C3 EM column shows Exact Match for metrics-generated (*): 31.9% vs. 2.4% ROUGE-L — a 13x discrepancy (§5.2).

Question Type	C0	C2	C3	C3 EM	C4	C5
metrics-generated	0.028	0.026	0.024	0.319*	0.024	0.017
novel-generated	0.261	0.257	0.258	0.025	0.269	0.254
domain-relevant	0.257	0.263	0.259	0.022	0.232	0.245

4.3 Results by Question Type

Metrics-generated ROUGE-L F1 is near-zero across all conditions (0.017–0.028), confirming the failure is technique-independent. The Exact Match rate on metrics-generated questions under C3 is 31.9% — 15 of 47 questions answered correctly — despite ROUGE-L = 0.024.

4.4 Cost and Latency

Since ROUGE-L F1 is approximately constant across measured conditions, cost and latency are the primary differentiators. C0 (12.3s) and C2 (11.2s) are the fastest. C3 adds 23 seconds of cross-encoder inference latency per query (35.3s total). C5 adds one extra GPT-4o call (13.2s). No enhanced condition lies on the efficient frontier.

5 Analysis

5.1 Structural Chunking Failure Is Technique-Independent

Financial tables in SEC filings have a characteristic structure: column headers at the top, row labels at the left, values in right-aligned numeric columns, spanning multiple pages. When PDF text is split at 512-token boundaries, these three components of a single data cell's meaning — column header, row label, value — are routinely placed in different chunks.

A similarity search for 'capital expenditure FY2018' retrieves the chunk containing the row label but not the chunk containing the value. The answer does not exist as a retrievable unit. No reranker can reorder chunks to surface a value that was never attached to its label. No query expansion can generate a hypothetical passage that matches a chunk of decontextualized numbers. No metadata filter can locate a passage that does not exist. This is why all five techniques fail at approximately equal rates on metrics-generated questions.

Finding 1: Structural chunking failure is technique-independent.

Standard RAG retrieval techniques cannot recover from structural chunking failure in financial tables. If the target value is separated from its row label and column headers during text splitting, no retrieval technique can assemble the complete passage from fragments. The fix requires table-aware document parsing that treats each table as a structured object, not a text stream. This is not a retrieval optimization; it is a preprocessing requirement.

The recursive chunking condition (C2) uses 2,000-token chunks specifically to preserve larger table sections, yet achieves metrics-generated ROUGE-L F1 = 0.026 — essentially the same as C0 (0.028). SEC filing tables frequently span more than 2,000 tokens when converted to text. Even a 4x larger chunk size does not recover financial table values. The fix is not larger chunks; it is table-aware parsing that preserves row-column structure independently of text splitting.

5.2 ROUGE-L Understates Performance on Numeric Questions

Under C3 (reranking), metrics-generated questions score ROUGE-L F1 = 0.024 but Exact Match = 0.319 — 15 of 47 numeric questions answered correctly despite near-zero ROUGE-L. This 13x discrepancy arises because GPT-4o answers numeric questions with step-by-step explanations. A response calculating '24.26' in 60 words achieves Exact Match = 1.0 (the string '24.26' appears) but ROUGE-L F1 $\sim$ 0.04 (minimal overlap with the gold answer '24.26').

Finding 2: ROUGE-L understates performance on numeric financial QA by 10–13x.

Exact Match reveals that GPT-4o correctly calculates numerical answers in 27.7–31.9% of metrics-generated questions under the best conditions, despite ROUGE-L scores of 0.017–0.028. We recommend that financial QA benchmarks report Exact Match alongside ROUGE-L. The implication: the model can reason numerically when given correct context; retrieval failure is the binding constraint, not reasoning failure.

5.3 The Efficient Frontier Collapses to a Point

The efficient frontier is the set of conditions not Pareto-dominated. Here it collapses to a point: C0 and C2 are the only non-dominated conditions. Reranking (C3) is strictly dominated: 2.9x latency at slight accuracy decrease. Query expansion (C5) is strictly dominated: higher cost at lower accuracy.

6 Ablation Study

6.1 Scope

Each of C1–C6 is an ablation of C0. This section gives component-level summaries.

6.2 A1 — Semantic Chunking (C1, pending)

Pending. Expected: no improvement on metrics-generated (same structural failure), possible minor improvement on novel-generated where semantic boundaries better preserve prose coherence.

6.3 A2 — Recursive Chunking (C2)

Matches C0 exactly at ROUGE-L F1 = 0.177. Metrics-generated: 0.026 (C0: 0.028). A 4x larger chunk size does not recover financial table values. Structural failure is not a function of chunk size.

6.4 A3 — Reranking (C3)

ROUGE-L F1 decreases from 0.177 to 0.175; overall Exact Match unchanged at 0.128; metrics-generated Exact Match increases from 0.277 to 0.319. The reranker surfaces correct chunks more often when they exist, but does not fix the structural case where correct chunks do not exist as complete units. Latency penalty: 23 seconds per query.

6.5 A4 — Metadata Enrichment (C4)

ROUGE-L F1 decreases from 0.177 to 0.170; Exact Match decreases from 0.128 to 0.105. The metadata prefix may be shifting query embeddings away from content-specific terms toward structural metadata terms, reducing semantic retrieval quality. This is reported rather than explained away; the mechanism is a hypothesis.

6.6 A5 — Query Expansion (C5)

ROUGE-L F1 decreases from 0.177 to 0.167; metrics-generated F1 decreases from 0.028 to 0.017 — the single largest degradation of any condition on any question type. Consistent with Patel et al. [3]. Hypothesized mechanism: GPT-4o's hypothetical answer contains plausible but incorrect numerical figures that retrieve passages with similar-but-wrong values.

7 Limitations

- **Single-run results only.** Three-seed replication is the immediate next step. Observed differences between conditions (0.000 to 0.010 below baseline in ROUGE-L F1) may be within single-run noise.
- **17 questions excluded.** Company investor pages that block automated downloads may differ systematically in document structure, introducing selection bias.
- **C1 and C6 pending.** EnsembleRetriever and ContextualCompressionRetriever moved to langchain_classic in LangChain 1.3.11; import path not resolved before submission.
- **all-MiniLM-L6-v2 may be the embedding ceiling.** Financial-domain-specific embeddings may change the relative condition ordering.
- **Cost-per-query not reported.** OpenAI usage logs available but not yet aggregated.
- **Table-aware parsing not tested.** Identified as the required fix; not implemented. Finding 3 is derived from failure analysis, not experimental evidence.
- **GPT-4o only.** Smaller or domain-specific models may produce different F1 values and condition orderings.
- **Hardware-specific latency.** Figures measured on Windows / Anaconda Python 3.11.

8 Discussion

8.1 What Practitioners Should Take Away

If a financial RAG system fails on questions like 'What was capex in FY2022?', adding reranking or query expansion will not fix it. The failure is that cash flow statement values are not reaching the model as coherent, contextually complete chunks. That requires parsing tables as structured objects with preserved row-column associations — a preprocessing step, not a retrieval optimization. Invest engineering effort in table-aware parsing before any retrieval optimization.

8.2 The Negative Result Is Not a Null Result

A negative result — no improvement — is not a null result — no effect. The finding rules out a large class of approaches and redirects effort toward the correct class. We report this directly. A paper that applied techniques known to work on general-domain benchmarks to financial documents and reported only positive conditions would misrepresent the problem.

8.3 Implications for Benchmark Design

Any financial QA benchmark reporting only ROUGE-L as its accuracy metric underestimates model capability on numeric questions by 10–13x in our experiment. Benchmark designers should report Exact Match alongside ROUGE-L, or decompose results by question type.

9 Future Work

Finding 3: Question-type routing is the most likely near-term path to improvement.

Routing qualitative questions (F1 $\sim$ 0.26) to the current RAG pipeline and numeric questions (F1 $\sim$ 0.028) to a dedicated table-extraction pipeline would outperform any single-pipeline approach. This is a recommendation derived from failure analysis, not a measured result — we have not implemented table-aware extraction.

- **Table-aware chunking:** treat each financial table as a structured object with preserved row-column associations. Highest-priority experimental condition identified by the results.
- **Three-seed replication:** establish variance estimates and determine whether observed differences are within single-run noise.
- **C1 and C6:** resolve the LangChain 1.3.11 import conflict and add semantic chunking and hybrid results.
- **Cost measurement:** aggregate OpenAI usage logs to per-condition cost-per-query figures.
- **Question-type routing:** route qualitative questions to the current pipeline and numeric questions to a table-extraction pipeline; measure whether this outperforms any single-pipeline condition.
- **Financial-domain embeddings:** replace all-MiniLM-L6-v2 with FinBERT-based encoders.
- **Full FinanceBench:** extend from 133 public questions to the full 10,231-question benchmark.

10 Conclusion

We present a controlled ablation study testing five standard RAG retrieval improvements on FinanceBench. None improve over the C0 baseline on ROUGE-L F1. The failure is structural: fixed-token chunking destroys table structure in SEC filings, and no retrieval technique — reranking, metadata enrichment, query expansion, or larger chunks — can recover table cell values that were never coherently chunked.

We show that ROUGE-L F1 understates model capability on numeric questions by 10–13x; Exact Match reveals 27.7–31.9% correct numerical answers despite near-zero ROUGE-L scores. The model can reason numerically when given the right context — retrieval is the binding constraint.

The practical recommendation is direct: use the cheapest pipeline (C0 baseline) and invest in table-aware document parsing before any retrieval optimization. Reranking adds 2.9x latency at no accuracy gain. Query expansion decreases accuracy while adding cost. The negative result is reported directly, limitations are enumerated explicitly, and all claims are scoped to what is implemented and measured.

References

[1] Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. FinanceBench: A New Benchmark for Financial Question Answering. *arXiv:2311.11944*, 2023.

[2] Natan Vidra et al. Improving Retrieval for RAG-based QA on Financial Documents. *arXiv:2404.07221*, 2024.

[3] A. Patel et al. From BM25 to Corrective RAG. *arXiv:2604.01733*, 2026.

[4] Zhiyu Chen et al. FinQA: A Dataset of Numerical Reasoning over Financial Data. In *Proceedings of EMNLP 2021*.

[5] FinAgentBench. *ACM ICAIF 2025*.

[6] FinTagging. *arXiv:2505.20650*, 2025.

[7] S. Banerjee et al. Metadata-Driven RAG for Financial QA. *arXiv:2510.24402*, 2025.

[8] J. Kim et al. Multi-Reranker for FinanceRAG. *arXiv:2411.16732*, 2025.

[9] FinGEAR. *EMNLP Findings 2025*.

[10] Greenback Bears. Finance is a Jungle. *arXiv:2411.07142*, 2024.